State Merit Aid Programs and College Major: A Focus on STEM

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Since 1991 more than two dozen states have adopted merit-based student financial aid programs, intended at least in part to increase the stock of human capital by improving the knowledge and skills of the state's workforce. At the same time, there has been growing concern that the United States is producing too few college graduates in science, technology, engineering, and mathematics (STEM) fields. Using microdata from the American Community Survey, this paper examines whether recently adopted state merit aid programs have affected college major decisions, with a focus on STEM fields. We find consistent evidence that state merit programs did in fact reduce the likelihood that a young person in the state will earn a STEM degree.

I. Introduction

State merit-based financial aid programs, such as Georgia's HOPE Scholarship, award scholarships to in-state students who meet some merit requirement based on high school grade point average (GPA; e.g., a 3.0 in Georgia) and sometimes SAT or ACT scores or class rank. State merit aid programs also require students to maintain a certain GPA in college to renew the award for subsequent years, although the required GPA differs

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across programs.[1] Award values per recipient vary widely, from about $600 to over $8,000 in 2009–10, as do grant recipients per full-time enrollment, which range from less than 2% to over 35%.[2] State merit aid programs have grown substantially since the early 1990s, and this growth has led to a sizable literature that explores the effects of these programs on college outcomes, including enrollment, persistence, completion, and post-college retention in the state.[3] One outcome that has received little attention is the effect of merit aid on the choice of college major, and, as we note below, what research has been conducted has limitations.[4] We explore the effect of merit-based aid on the choice of majors using American Community Survey (ACS) data, with a particular focus on science, technology, engineering, and mathematics (STEM) majors.

[1] The minimum GPA for renewal is typically 2.75–3.0, but it is as low as 2.3 for the first year in Louisiana and 2.5 in subsequent years. A few states also have requirements that students must be continuously enrolled and must take a minimum number of credits.

[2] Brookings Institution, "Inventory of State Grants," available at http://www.brookings.edu/research/reports/2012/05/08-grants-chingos-whitehurst; accessed July 20, 2012.

[3] Researchers estimating the effects of state merit aid programs on college enrollment include Dynarski (2000, 2004), Cornwell, Mustard, and Sridhar (2006), Singell, Waddell, and Curs (2006), Goodman (2008), Farrell and Kienzl (2009), Orsuwan and Heck (2009), Zhang and Ness (2010), Winters (2012), Hawley and Rork (2013), and Bruce and Carruthers (2014). This literature typically finds a significant positive effect on the probability of attending college in state but inconsistent effects on the overall probability of attending college. Studies of merit aid effects on persistence and completion include Henry, Rubenstein, and Bugler (2004), Dynarski (2008), Scott-Clayton (2011), and Sjoquist and Winters (2012, forthcoming). Dynarski, Henry et al., and Scott-Clayton find a significant positive effect on the graduation rate, while Sjoquist and Winters and Fitzpatrick and Jones find no significantly positive effects. Post-college in-state retention has been explored by Hickman (2009), Fitzpatrick and Jones (2012), Hawley and Rork (2013), and Sjoquist and Winters (2013, 2014). Hu, Trengove, and Zhang (2012) review this literature.

[4] There is a sizable literature that explores the effect of various factors on the choice of college major, particularly STEM majors. See Delaney (2007), Crisp, Nora, and Taggart (2009), and Wang (2012) for surveys of the literature. Research among economists on the choice of college major dates back to at least Freeman (1971). In a life cycle framework, the choice of major depends on student preferences, the cost of completing the major, the student's ability (i.e., the probability of successfully completing the major course of study), and the expected earnings after graduation (Berger 1988). Montmarquet et al. (2002) extend the model to include uncertainty regarding the successful completing of each major (see also Arcidiacono 2004; Griffith 2010; Arcidiacono, Hotz, and Kang 2012; Beffy, Fougere, and Maurel 2012; and Wiswall and Zafar 2015). Others have considered the effect of academic ability and preparation (see Federman 2007; Griffith 2010; Kokkelenberg and Sinha 2010; Ost 2010; Rask 2010; and Stinebrickner and Stinebrickner 2014).

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Understanding the effect of merit-based aid on the choice of major is of interest for at least two reasons. First, the choice of major has a significant effect on a student's post-college earnings (see Arcidiacono [2004] and the references therein). Second, from a policy perspective, there is a significant concern that the United States is not producing a sufficient number of majors in STEM fields (National Academies 2010). STEM fields are major drivers of innovation and therefore have important consequences for long-run economic growth, national defense, and individual welfare (Atkinson and Mayo 2010). Thus, knowing whether merit-based aid has an unintended consequence of reducing STEM majors is relevant to the national focus on STEM majors.

We identified four empirical papers that consider the effects of financial aid on the choice of college major, none of which examine a national microdata sample like the ACS. First, Stater (2011) argues that increased financial aid lowers the price of majors that offer current consumption benefits and encourages student substitution toward such majors. Similarly, financial aid could be viewed as a transitory income shock leading to more current-consumption-oriented majors.[5] Stater (2011) considers the effect of student aid on the choice of the first-year major using student data from three large public universities. He finds that loans and grants have small and generally insignificant effects, while merit aid increases the probability of declaring a major in the humanities and in science and reduces the probability of majoring in social sciences. While Stater controls for student characteristics, he is unable to control for the endogeneity of aid. Since merit aid is received by students with better academic backgrounds, his merit aid variable could reflect unmeasured student characteristics that affect choice of major.

Rothstein and Rouse (2011) suggest that student loan debt might affect a student's choice of college major and future occupation due to debt aversion and credit constraints. Financial aid should decrease student loan debt and may reduce the importance of future earnings in college major decisions. Rothstein and Rouse (2011) explore their supposition using student records from an unidentified, wealthy, highly selective university that adopted a policy that replaced its student loan program with a student grant program. Their difference-in-differences regressions for choice of major yield small and generally statistically insignificant coefficients.

Cornwell, Lee, and Mustard (2008) explore the effect of Georgia's HOPE program on choice of major among freshmen enrolled at the University of Georgia. They find that HOPE had a statistically significant

[5] Riegle-Crumb et al. (2012), using the National Education Longitudinal Study (NELS) and High School and Beyond (HS&B), find that an additional $10,000 of family income reduces the probability that a student will declare a physical science/engineering major by 0.2–0.5 percentage points. However, permanent family income may have very different effects than transitory income from financial aid.

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positive effect on the probability of being an education major but no effect on other majors. Cornwell, Lee, and Mustard (2005, 2008) suggest that the requirement that students maintain a 3.0 GPA to retain the HOPE Scholarship in Georgia could cause some students to engage in strategic behaviors to avoid losing HOPE.[6] This could include assuming lighter course loads and even majoring in a subject for which it is easier to maintain the required GPA.[7]

Zhang (2011) investigates the effect of the Florida and Georgia merit-based aid programs on STEM majors using annual statewide totals of STEM degrees conferred obtained from the Integrated Postsecondary Education Data System's (IPEDS) Completion Survey. He finds no statistically significant effect of merit-based aid on the percentage of STEM majors, except for a 1.6 percentage point increase for private schools in Florida. Using aggregate data on degree conferrals is problematic because merit programs affect where students attend college and have been shown to increase the average quality of students who stay in state for college. Student quality also affects the choice of a major and needs to be taken into account by using microdata.

There are other possible mechanisms through which merit aid could affect the choice of a major. For example, high school GPA eligibility requirements for merit scholarships may provide incentives for students to take easier courses in high school, which could make them less prepared for more challenging majors in college; alternatively, if merit programs increase student effort in high school, as suggested by Henry and Rubenstein (2002), they could cause students to be better prepared for college. In addition, Dynarski (2000) and Cornwell, Mustard, and Sridhar (2006) find that merit programs increase the likelihood that students stay in state for college and affect the type of in-state institution they attend, which could alter the college major options available and alter the relative attractiveness of various majors. In particular, merit programs increase the percentage of high-ability students who stay in state, which may make it harder for more-moderate-ability students to obtain admission to the most selective institutions in the state, which could affect these students' choice of major. Furthermore, if grading is done on a relative basis, merit-induced grade competition from better students may cause moderate-ability students to earn lower grades in difficult majors and encourage

[6] This could be especially important if students are loss averse or engage in hyperbolic discounting. Carruthers and Özek (2012) also suggest that losing merit aid may cause students to increase hours worked in paid employment.

[7] Dee and Jackson (1999) find that students in science, engineering, and computing are substantially more likely to lose the Georgia HOPE scholarship than students in other disciplines, but they do not consider the effects that this might have on students' major choices.

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them to switch to easier majors to stay in college.[8] Similarly, institutions experiencing increased enrollment due to merit aid may raise standards in STEM fields to push lower-ability students into other majors with lower instructional costs.

We investigate the effect of merit-based financial aid on student choice of college major, focusing especially on STEM majors, using the American Community Survey (ACS). Unlike Zhang (2011), who considers merit aid in only two states, we consider the effects in the 27 states that adopted merit-based aid programs between 1991 and 2005, although we focus on the nine states that have more significant merit aid programs. More importantly, our ACS analysis is unaffected by merit-induced changes in student quality because our merit variable is measured based on state of birth and not state of college attendance.

To preview our results, we find a consistent story that significant merit-based aid programs reduce the probability that a young person will complete a bachelor's degree with a STEM major. This is a new and important result in the nascent literature. Using ACS data, our baseline specification suggests that adopting a strong merit aid program reduces the number of STEM graduates from the state by 6.5%. Accounting for measurement error suggests that the decrease could be as large as 9.1%.

II. Data and Empirical Framework

Our interest is in whether merit-based aid affects a student's choice of major, focusing on STEM majors but also considering the effects on other majors. We use the list of majors that the US Immigration and Customs Enforcement uses to define a STEM major. Beginning in 2009, the Census Bureau asked respondents to specify their college major as part of the ACS, and thus we can use ACS data for 2009–11 for the analysis. Appendix table A1 reports the match of ACS college majors to the broad major categories we use. We follow the basic approach used by others who have used Census Bureau data to investigate the effect of merit-based aid on various student outcomes (see Dynarski 2008; Hickman 2009; and Sjoquist and Winters 2012) and thus provide only a brief explanation below of the data and the empirical approach.

We follow a treatment and control research design implemented via a difference-in-differences regression framework. The treatment group consists of individuals who were exposed to a broad-based state merit aid program when they graduated from high school, and the control group

[8] Arcidiacono, Aucejo, and Hotz (2013) and Luppino and Sander (2015) report evidence of peer effects using data from the University of California system. Attending a University of California campus where the sciences are more competitive makes a moderate-ability student less likely to earn a science degree.

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consists of individuals who were not. Unfortunately, we do not know when and where individuals graduated from high school, so we follow previous literature (e.g., Dynarski 2008; Hickman 2009; and Sjoquist and Winters 2012) and assign individuals to the merit program treatment group based on state of birth and year of birth, where year of birth is computed as the year of the survey minus age at the time of the survey. If an individual was born in a state that adopted a merit program and turned 18 after the program was implemented, they are assigned to the treatment group; if not, they are part of the control group. The control group, therefore, includes both individuals born in states that never adopted merit programs and individuals born in merit-adopting states but turning 18 before the program was implemented. Because some individuals attend high school outside their state of birth and some do not graduate from high school at age 18, our treatment assignment will have some degree of measurement error. We take up the issue of measurement error in more detail later.

Based on various sources (see the sources note to table 1), we identified 27 states that implemented a merit-based student financial aid program between 1991 and 2005, although the characteristics of these programs differ substantially. Some of these state merit aid programs are relatively small and not likely to have a sizable impact on education outcomes. After reviewing the details of all of these programs, we identified nine merit aid programs as being relatively more significant as measured by the eligibility criteria, the number of students in the program, and the size of the award. Table 1 lists characteristics of these nine "strong" merit aid states (hereafter, "strong merit states") and also lists in its note the 18 states adopting "weak" merit aid programs (hereafter, "weak merit states").

We restrict the sample to persons who have at least a bachelor's degree and are between ages 24 and 39, so that every strong merit state has multiple cohorts both before and after the merit aid program was adopted. We restrict the sample to those with at least a bachelor's degree, since a major is not reported for those who have not completed college. If merit aid affected completion, we could have an endogenous selection problem. However, as we show in Sjoquist and Winters (forthcoming) and below, merit aid has no meaningfully positive effect on college completion. We also exclude individuals with imputed information for age, education, college major, or state of birth, but results are robust to including these individuals.

Our preferred specification for assessing the effects of broad-based merit scholarship programs on STEM majors is to compare states with strong merit programs to states with no merit program; the District of Columbia is included as a state. We do, however, examine the robustness of our results to including weak merit states in the analysis.

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Table 1
States Adopting Strong Merit Aid Programs

Program Name	First Cohort	Initial Requirement	Renewal GPA	Award Amount	Award per Recipient, 2010 ($)	Recipients as a Percent of Undergraduates, 2010 (%)
Florida Bright Futures Scholarship	1997	3.0–3.5 high school GPA and 970–1270 SAT/20–28 ACT	2.75–3.00	75%–100% of tuition and fees	2,381	31.9
Georgia HOPE Scholarship	1993	3.0 high school GPA	3.00	Tuition and fees	3,877	43.0
Kentucky Educational Excellence Scholarship	1999	2.5–4.0 high school GPA plus ACT bonus	2.50–3.00	$500–$3,000	1,381	44.7
Louisiana TOPS Scholarship	1998	2.5 high school GPA and 20 ACT	2.30–2.50	Tuition and fees	3,050	22.6
Nevada Millennium Scholarship	2000	3.0 high school GPA	2.60–2.75	$80 per credit	1,279	23.8
New Mexico Lottery Success Scholarship	1997	2.5 first-semester college GPA	2.50	Tuition and fees	2,388	19.6
South Carolina LIFE Scholarship	1998	3.0 high school GPA and 1100 SAT/24 ACT	3.00	$5,000-$7,500	4,675	23.2
Tennessee HOPE Scholarship	2003	3.0 high school GPA or 1000 SAT/21 ACT	2.75–3.00	$2,500-$4,000	3,423	33.5
West Virginia PROMISE Scholarship	2002	3.0 high school GPA and 1000 SAT/21 ACT	2.75–3.00	Tuition and fees	4,943	12.1

SOURCES.—Dynarski (2004), Heller (2004), the Brookings Institution, and state agency websites.

NOTE.—Eighteen other states adopted "weak" merit aid programs. These include Alaska, Arkansas, California, Idaho, Illinois, Maryland, Massachusetts, Michigan, Mississippi, Missouri, Montana, New Jersey, New York, North Dakota, Oklahoma, South Dakota, Utah, and Washington. For several states the renewal GPA increases after the first renewal point, and hence the range given.

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We estimate linear probability models (LPM) as follows:

$$P(Y_{isct} = 1) = \Gamma_s + \Pi_c + \beta X_{isct} + \delta_s T_{sc} + \theta \text{Merit}_{sc} + \varepsilon_{isct},$$

where Y equals one if the individual completed a bachelor's degree in a STEM field (either the first or second major), i denotes an individual, s denotes the state of birth, c denotes the year-of-birth cohort, t denotes the survey year, Γ_s includes state-of-birth fixed effects, Π_c includes year-of-birth cohort fixed effects, X includes dummy variables for individual characteristics, including sex, race, Hispanic origin, and age, T_{sc} is state-of-birth by year-of-birth time trends, and Merit is an indicator variable equal to one if the individual was exposed to a state merit program and zero otherwise. The state-of-birth and year-of-birth fixed effects allow the model to be interpreted as a difference-in-differences model identified by differences across birth states and across birth cohorts within birth states. We present results for linear models since the coefficients are easier to interpret, but results are robust to using probit or logit. Summary statistics are reported in table 2 for the full sample and separately for females and males.

Because we use individual-level data and the Merit variable is defined based on state and year of birth, ordinary least squares (OLS) standard errors should not be used because they do not account for intracluster correlation (Bertrand, Duflo, and Mullainathan 2004; Donald and Lang 2007; Cameron, Gelbach and Miller 2008). Instead, we report both stan-

Table 2
Summary Statistics for ACS Data

	All Graduates		Females		Males	
	Mean	SD	Mean	SD	Mean	SD
STEM major	.206	.405	.137	.344	.294	.455
Business major	.204	.403	.171	.377	.247	.431
Education major	.108	.310	.151	.358	.052	.223
Health major	.056	.230	.086	.280	.018	.133
Liberal arts major	.237	.425	.243	.429	.228	.420
Social science major	.189	.391	.211	.408	.160	.367
Strong merit dummy	.056	.230	.057	.233	.054	.226
Weak merit dummy	.171	.376	.174	.379	.166	.372
Age	31.425	4.597	31.336	4.592	31.539	4.601
Female	.559	.496	1.000	.000	.000	.000
Black	.079	.269	.090	.286	.064	.246
Hispanic	.061	.239	.064	.244	.057	.231
Asian	.031	.173	.029	.167	.034	.180
Other nonwhite	.021	.143	.022	.145	.020	.140
Total no. of observations	418,739		238,395		180,344	

NOTE.—The sample includes all US-born college graduates between ages 24 and 39 in the 2009–11 American Community Survey with unimputed information for age, education, college major, and birth state.

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dard errors clustered by state of birth and 95% confidence intervals based on procedures suggested by Conley and Taber (2011). Clustered standard errors are typically preferred to OLS standard errors, but Conley and Taber (2011) show that clustered standard errors can be downwardly biased when the number of policy changes is small. They suggest a confidence interval procedure based on the distribution of residuals across the control states and show that their procedure outperforms conventional clustered standard errors when there are a small number of treatment groups and that it does no worse more generally.[9] Our preferred specification includes nine states with policy changes, so we report both clustered standard errors and Conley-Taber confidence intervals.

III. Empirical Results

A. Basic Results

Table 3 reports our basic results; we report results for the entire sample and separately for females and males. All regressions include dummies for state of birth, year of birth, age, sex (except for the gender-specific regressions), and race/ethnicity, and state-of-birth by year-of-birth time trends. We include state-of-birth by year-of-birth time trends to account for the possibility of differential trends in college majors across states. Our preferred specification is in column 1, in which the treatment group includes only persons born in strong merit states who were exposed to a strong merit program; persons born in weak merit states are excluded from both the treated and control states in column 1. The control group thus includes persons born in non–merit states and persons born in strong merit states but reaching age 18 before the merit program was implemented. The remaining columns in table 3 explore the sensitivity of our results to using different treatment and control states. It is not clear how weak merit states should be considered. They could be considered treated states, but they only receive a partial treatment compared to the strong

[9] The Conley-Taber inference procedure imposes fairly minimal assumptions. In addition to the linearity, exogeneity, and full rank assumptions of the linear probability model, their procedure only assumes that the state-by-year errors are independent and identically distributed across states and have a bounded density. Their inference procedure computes mean residuals by state and year and then estimates the cumulative distribution function for the treatment effect estimator based on the empirical distribution of the mean residuals in the control states. Assuming there are N treatment states, this amounts to randomly drawing N pseudo-treatment states from the combined group of treatment and control states, estimating a pseudo-treatment effect, and doing this thousands of times. A confidence interval is then constructed based on the appropriate quantiles of the empirical distribution. The actual treatment effect estimate is then compared to the distribution of pseudo-treatment effects, and we reject the null of zero treatment effect if the actual treatment effect estimate is outside the confidence interval.

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Table 3
Merit Program Effects on STEM Majors for Different Treatment and Control States

	(1)	(2)	(3)	(4)	(5)
A. Total population	−.0133	−.0126	−.0122	−.0135	−.0051
	(.0053)**	(.0055)**	(.0048)**	(.0066)*	(.0047)
	[−.0261, −.0049]	[−.0237, −.0036]	[NA]	[NA]	[−.0168, .0034]
	{−6.45%}	{−6.11%}	{−5.92%}	{−6.55%}	{−2.49%}
B. Females	−.0057	−.0060	−.0056	−.0055	.0016
	(.0093)	(.0091)	(.0098)	(.0103)	(.0042)
	[−.0172, .0090]	[−.0162, .0072]	[NA]	[NA]	[−.0087, .0124]
	{−4.22%}	{−4.44%}	{−4.14%}	{−4.07%}	{1.18%}
C. Males	−.0240	−.0220	−.0216	−.0244	−.0138
	(.0099)**	(.0088)**	(.0101)**	(.0079)**	(.0086)
	[−.0468, −.0104]	[−.0400, −.0101]	[NA]	[NA]	[−.0317, .0012]
	{−8.05%}	{−7.38%}	{−7.25%}	{−8.19%}	{−4.75%}
State-of-birth dummies	Yes	Yes	Yes	Yes	Yes
Year-of-birth dummies	Yes	Yes	Yes	Yes	Yes
Age dummies	Yes	Yes	Yes	Yes	Yes
Sex, race/ethnicity dummies	Yes	Yes	Yes	Yes	Yes
State-of-birth × year-of-birth trends	Yes	Yes	Yes	Yes	Yes
Region × year-of-birth dummies	No	No	Yes	No	No
Strong merit states	Treatment	Treatment	Treatment	Treatment	Treatment
Weak merit states	Excluded	Control	Excluded	Excluded	Treatment
Non–merit states	Control	Control	Control	Excluded	Control
Total states	33	51	33	9	51

NOTE.—Standard errors in parentheses are clustered by state of birth. Conley-Taber 95% confidence intervals are in squared brackets; these are inestimable for columns 3 and 4 because of how the control group is specified. The numbers in braces are the coefficient divided by the corresponding pre-merit mean for the STEM dummy and represent the percentage change in the number of STEM graduates from the state.

* $p < .10$ based on standard errors clustered by state of birth.

** $p < .05$ based on standard errors clustered by state of birth.

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merit states, and treating the full and partial treatment states the same would likely attenuate coefficient estimates. Alternatively, we could treat weak merit aid states as control states, but that would also likely attenuate coefficient estimates. We consider both as alternatives given the uncertainty of how they should be considered, but our preferred estimates exclude the weak merit states.

In the second column of table 3, persons born in the weak merit aid states are included as part of the control group. The third column modifies column 1 by including region by year-of-birth dummies; this means that the comparison group includes only states in the same census region. However, strong merit programs were only implemented in the South and West regions, so column 3 does not fully utilize information for the Northeast and Midwest regions, and Conley-Taber confidence intervals are inestimable because we have too few control states. The fourth column drops all weak merit and non–merit states so that the analysis is simply a pre– and post–merit aid comparison among strong merit aid states; that is, the control group includes only persons born in strong merit states but turning age 18 before the program in their state was implemented. The fifth column includes all 51 states in the sample and includes persons exposed to weak merit aid programs as part of the treatment group.[10]

The results in columns 1–4 of table 3 tell a consistent story, namely, that a strong merit aid program reduces the probability that an individual graduates from college with a STEM major, with the coefficients on the merit variable having similar magnitude across the first four columns.[11] The results imply that a strong merit aid program reduces the probability of graduating with a STEM major by about 1.3 percentage points. Given that only one-fifth of all college graduates major in a STEM field, the magnitude of the effect is quite large. The results suggest that adopting a strong merit aid program reduces the number of STEM graduates by 6.5%.

[10] Our ACS data do not allow us to control for student characteristics, and in particular for changes in student quality, which might affect choice of major. However, we do not expect that changes in student quality due to factors other than merit aid would systematically differ over time between merit and non–merit aid states. To the extent that merit aid increased measured student quality, which in turn affected the choice of majors, this then is a possible mechanism through which merit aid affects choice of major.

[11] It turns out that the confidence intervals from clustered standard errors and the Conley-Taber procedure are fairly similar for our baseline specification with 9 treatment and 24 control states, though one is sometimes larger than the other and the Conley-Taber intervals are by construction not perfectly symmetric. This indicates that the clustered standard errors actually do a reasonably good job in this setting, but we still feel it useful to report both inference methods. With fewer treatment states, the clustered standard errors would begin to perform much worse, and it is unclear exactly how many treatment states is "enough" to produce reliable inferences via clustered standard errors.

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It is well known that the choice of major differs by gender, and thus we estimated the regressions separately for females and males (second and third panels in table 3). The results imply that it is males, and not females, who experience a decrease in the probability of earning a STEM major as the result of the adoption of a strong merit aid program. While the coefficient on merit aid is negative for females, it is not statistically significantly different from zero; however, the Conley-Tabor confidence intervals do not rule out the possibility of substantive economic effects.[12] A potential explanation for the gender difference is that females may be less sensitive to the provision of merit aid, which would be consistent with the finding that choice of major by females is less sensitive to future net earnings (Montmarquette, Canning, and Makserdjian 2002; Freeman and Hirsch 2008; Wiswall and Zafar 2015). The differential findings by gender are also consistent with findings in Griffith (2010) and Rask (2010) that decisions to major in the sciences are more sensitive to grades for males than for females. Thus, merit aid may have a greater effect on males than females because males are more likely to switch away from STEM to increase their chances of keeping their merit scholarship.

In column 5 we include the weak merit aid states as part of the treated states. For this regression, the coefficients on merit aid are still negative for the entire sample and for males, but they are much smaller in magnitude and not statistically significant, implying that weak merit aid programs have only a small effect on the probability that students will major in a STEM field. To further explore this, we ran a regression similar to that of column 1 (but not reported here) except that we excluded the strong merit aid states and used only the weak merit aid states as the treated states. The coefficient estimates are small and statistically insignificant.[13] We also estimated regressions (but not reported here) using a dummy variable for strong merit aid states and a dummy variable for weak merit aid states. The coefficients and standard errors for the strong merit aid dummy are very similar in magnitude to those reported in column 1 of table 3. The coefficients on the weak merit aid dummies are relatively small and statistically insignificant. In addition, we explored the effect on the coefficients in column 1 of table 3 of changing which states are considered strong merit aid states. In the first regression, we included Michigan and Mississippi as

[12] Zhang (2011) also finds differential effects of merit aid by gender.

[13] While we list California as having a merit aid program, the California program requires only a 2.5 GPA (grade point average) and has a maximum income threshold. Thus, one might question whether California should be considered even a weak merit aid state. We reestimated col. 5 but dropped California from the analysis. The resulting coefficient on merit aid is −0.0095 and is statistically significant. Therefore, excluding California increases the coefficient in col. 5 but it is still smaller than the coefficient in col. 1, confirming that weak merit programs have weaker effects than the strong merit programs.

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strong merit aid states, and in a second regression we also included Arkansas and Maryland to the strong merit aid states; these are four of the strongest of the weak merit programs and could be considered "strong" programs, as in Dynarkski (2004). In a third regression, we excluded Kentucky and Nevada, which could be considered two of the weaker nine strong merit states, from the strong merit group. The results (but not reported here) are very similar to those in column 1 of table 3, suggesting that our main results are insensitive to small reasonable variations in which states are included in the treatment.

We also examined the effects of merit programs on STEM majors separately for each individual state. Except for Nevada and Tennessee, we found negative coefficients that ranged from −0.0054 to −0.0631 for all of the strong merit states. However, the individual coefficients are noisily estimated, so we do not report them.

B. Alternative Time Trend Specification

The results thus far assume state-of-birth × year-of-birth linear trends. We also estimated column 1 of table 3 excluding the state-of-birth × year-of-birth trends. The results are qualitatively the same, although the coefficients are a little smaller in magnitude than in column 1 of table 3, −0.0093 for the total population. However, our inspection of the raw means revealed a pre–merit program upward trend in the percentage of college graduates majoring in STEM in strong merit states; results that ignore this trend are likely inaccurate.

One alternative to our baseline assumptions is to conduct an event style analysis in which we allow the effects of the merit programs to vary over time by including separate merit indicators for each year before and after the policy in event time. This would also embed a placebo test since there should be no effect in the years immediately before the policy.[14] We conducted such an analysis (results not shown). The results are qualitatively consistent with our baseline results but are quite noisy and are not reported. Instead, we report results in which we include an event time trend variable and the trend variable interacted with the strong merit aid dummy. The event time trend measures years from the date of the adoption of the merit aid program, with negative values for the pre–adoption

[14] In results not shown, we also conducted explicit placebo tests by pretending that the states adopting strong merit programs did so 5 years before they actually adopted the programs and estimating the effects of these fake treatments excluding those who were actually treated. Estimating a placebo regression with the specification in col. 1 of table 3 gives a small coefficient of −0.0026 that is statistically insignificant. Similar null results for placebo tests are found when we vary the placebo adoption year, implying that the significant negative effects we find for the actual treatment are unlikely to have occurred for reasons other than the treatment, strengthening our confidence in the results for the actual treatment.

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period and positive for the post–adoption period. We then estimate regressions under the same specifications in column 1 of table 3, except that we do not include the state-of-birth × year-of-birth trends since this exercise is intended as an alternative to state-specific linear trends. Results are reported in table 4.

The coefficients on the trend variable are positive but statistically significant only for males. The positive coefficient implies an upward prepolicy trend in the probability of earning a degree with a STEM major in strong merit states. The coefficients on the interaction term are negative and statistically significant for the total population and males, implying that the upward trend reversed after adoption of merit aid. Even with this alternative time trend specification for strong merit states, the coefficient of −0.0121 on the strong merit dummy for the total population is still qualitatively similar to the baseline estimate.[15]

C. Non-STEM Majors

Although our focus is on STEM majors, we also consider the effect of merit aid on other majors, with the results presented in table 5 using our preferred specification. We first combined all of the non-STEM majors into five broad categories: business, education, health, liberal arts, and social sciences; summary statistics for these are in table 2.[16] Given that strong merit aid programs reduce the probability that an individual majors in a STEM field, there must be an increase in the likelihood of majoring in a non-STEM field. Furthermore, examining the effects on other majors may help us understand why merit programs discourage students from being a STEM major.

In a life cycle framework the choice of major depends on the expected earnings after graduation and the cost of completing the major, given the student's ability and preferences (Berger 1988). Before presenting the regression results, it is useful to consider earnings and cost. Consider, first, mean earnings for each broad major. We compute national mean earnings for persons ages 40–49 since they were not exposed to merit programs; mid-career earnings may also be more relevant than starting salaries (Berger 1988).

[15] Another alternative time-trend specification is to include separate pre- and post-merit trends for each individual merit state instead of the joint trends in table 4. Doing so gives a merit coefficient of −0.0132, which is very similar to our baseline estimate.

[16] Note from appendix table A1 that we define business, liberal arts, and social sciences very broadly to minimize the number of groups. The results for these variables are generally robust to marginal changes in their definitions. We also compared the college major means in table 2 to means computed from the Integrated Postsecondary Education Data System (IPEDS). Although the ACS and the IPEDS are not directly comparable, we did find the means to be fairly similar.

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Table 4
Merit Program Effects on STEM Majors Using Pre- and Post-Merit Event Time Trends

	Total (1)	Females (2)	Males (3)
Strong merit dummy	−.0121	−.0034	−.0239
	(.0058)**	(.0108)	(.0085)***
	[−.0241, −.0027]	[−.0201, .0062]	[−.0388, −.0026]
	{−5.87%}	{−2.52%}	{−8.02%}
Event time trend	.0014	.0007	.0023
	(.0009)	(.0012)	(.0010)**
Event time × strong merit dummy	−.0023	−.0011	−.0037
	(.0012)*	(.0021)	(.0013)***
State-of-birth dummies	Yes	Yes	Yes
Year-of-birth dummies	Yes	Yes	Yes
Age dummies	Yes	Yes	Yes
Sex, race/ethnicity dummies	Yes	Yes	Yes
State-of-birth year trends	No	No	No
Strong merit states	Treatment	Treatment	Treatment
Weak merit states	Excluded	Excluded	Excluded
Non–merit states	Control	Control	Control
Total states	33	33	33

NOTE.—Standard errors in parentheses are clustered by state of birth. Conley-Taber 95% confidence intervals are in squared brackets. The numbers in braces are the coefficient divided by the corresponding pre-merit mean for the STEM dummy and represent the percentage decrease in the number of STEM graduates from the state.

* $p < .10$ based on standard errors clustered by state of birth.

** $p < .05$ based on standard errors clustered by state of birth.

*** $p < .01$ based on standard errors clustered by state of birth.

Mean earnings in 2011 dollars are $95,389 for STEM, $78,122 for business, $67,735 for social sciences, $58,937 for health, $58,823 for liberal arts, and $46,169 for education. While out-of-pocket costs for textbooks, lab fees, and so forth, do vary by major, the primary additional cost of a STEM major is thought to be the additional time and effort that is required in STEM courses. Arcidiacono, Aucejo, and Spenner (2012) show that at Duke University natural science, engineering, and economics courses are more difficult, are associated with more study time, and have harsher grading standards. There are several studies that find that grades are lower for courses or majors in STEM fields. For example, Betts and Morell (1999) find that students at the University of California, San Diego, who majored in engineering or science had lower GPAs than students in other majors, controlling for student ability, socioeconomic variables, and resources at the student's high school. Rask (2010) reports that average grades in science and math courses at an unnamed northeastern liberal arts college are generally lower and have a wider distribution than grades in other courses.

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Table 5
Merit Program Effects on Other Broad Majors

	Business (1)	Education (2)	Health-Related (3)	Liberal Arts (4)	Social Sciences (5)	Major "Quality" (6)
A. Total population	.0057	−.0110	.0003	.0124	.0058	−.0065
	(.0077)	(.0062)*	(.0054)	(.0049)**	(.0031)*	(.0032)*
	[−.0043, .0160]	[−.0172, .0010]	[−.0073, .0070]	[.0007, .0218]	[−.0031, .0176]	[−.0123, −.0023]
	{2.66%}	{-8.12%}	{.42%}	{6.21%}	{3.36%}	
B. Females	.0133	−.0145	.0023	−.0083	.0128	.0001
	(.0152)	(.0107)	(.0097)	(.0088)	(.0068)*	(.0043)
	[−.0027, .0258]	[−.0261, .0004]	[−.0097, .0146]	[−.0212, .0071]	[−.0005, .0253]	[−.0057, .0062]
	{7.48%}	{−7.57%}	{2.14%}	{−4.23%}	{6.67%}	
C. Males	−.0040	−.0057	−.0022	.0400	−.0040	−.0132
	(.0070)	(.0060)	(.0049)	(.0085)***	(.0103)	(.0045)***
	[−.0209, .0123]	[−.0131, .0035]	[−.0056, .0049]	[.0200, .0558]	[−.0186, .0126]	[−.0216, −.0069]
	{−1.53%}	{−8.97%}	{−8.88%}	{19.57%}	{−2.71%}	
State-of-birth dummies	Yes	Yes	Yes	Yes	Yes	Yes
Year-of-birth dummies	Yes	Yes	Yes	Yes	Yes	Yes
Age dummies	Yes	Yes	Yes	Yes	Yes	Yes
Sex, race/ethnicity dummies	Yes	Yes	Yes	Yes	Yes	Yes
State-of-birth trends	Yes	Yes	Yes	Yes	Yes	Yes
Strong merit states	Treatment	Treatment	Treatment	Treatment	Treatment	Treatment
Weak merit states	Excluded	Excluded	Excluded	Excluded	Excluded	Excluded
Non–merit states	Control	Control	Control	Control	Control	Control
Total states	33	33	33	33	33	33

NOTE.—Standard errors in parentheses are clustered by state of birth. Conley-Taber 95% confidence intervals are in squared brackets. Major "quality" is measured using regression-adjusted major-specific earnings of persons ages 40–49 in the American Community Survey. The numbers in braces are the coefficient divided by the corresponding pre-merit mean for the major dummy and represent the percentage change in the number of major graduates from the state.

* $p < .10$ based on standard errors clustered by state of birth.

** $p < .05$ based on standard errors clustered by state of birth.

*** $p < .01$ based on standard errors clustered by state of birth.

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Consider, first, the results in columns 1–5 of table 5, where each regression is separately estimated using our baseline specification. For the total population, we find that other than for education, the coefficients on merit aid are positive but statistically significant only for liberal arts and social sciences. The results by gender suggest that the effect on the social science major is driven by the reaction of females to merit aid, while the effect on liberal arts is driven by the response of males. The coefficient for the education major is negative and statistically significant for the total population but statistically insignificant in the gender-specific regressions. The negative coefficient for education is contrary to the expectations implied by Cornwell, Lee, and Mustard (2008), who consider the declared freshman major. Interestingly, these results are not strictly consistent with a shift away from high-paying majors and toward low-paying majors, as suggested by the mechanisms of Rothstein and Rouse (2011) and Stater (2011). Finding no shift away from business is particularly noteworthy given its relatively high salary.

We next created a new dependent variable denoted "major quality," based on major-specific earnings, with the results reported in column 6 of table 5. To construct this variable we computed regression-adjusted mean log earnings for workers ages 40–49 for 176 individual majors in the ACS. We then use regression-adjusted mean log earnings as a measure of major quality based on the assumption that higher "quality" majors have higher earnings. Future earnings are only one dimension of major quality, but they are arguably one of the more important dimensions. The coefficients in column 6 are negative and statistically significant for the total population and for males, and they are virtually zero and insignificant for females. The implication is that strong merit aid programs on average shift males from higher to lower earning college majors, resulting in 1.3% lower earnings. This is a not a huge effect, but it is certainly nontrivial.

D. Robustness

We considered several robustness checks (results are not shown here). We first estimated the equation in column 2 of table 3 but included the region by year-of-birth dummies. The results are essentially the same as those in column 2. We also estimated column 1 but included several additional state-of-birth control variables, measured the year an individual was age 18:[17] cohort size, the state unemployment rate, median household income, the share of the population that is foreign born, mean K–12 pupil-teacher ratio, mean public K–12 expenditures per pupil, and mean SAT scores. It is possible that there are constraints on the possible number of STEM majors, so larger cohorts might negatively affect the probability of being a STEM major. There is some evidence that the mix of majors is

[17] Results are robust to instead measuring these variables at age 20.

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related to economic cycles (Bradley 2012), so we included the unemployment rate. In addition, as noted above, the likelihood that someone might major in a STEM field has been found to be negatively related to household income. Orrenius and Zavodny (2015) find that the foreign born share decreases the likelihood that natives major in STEM fields. Finally, education inputs could have changed and altered student preparation for college.[18] The results from including these additional control variables are essentially the same as those in column 1.

The results in table 3 are conditioned on persons earning a bachelor's degree. But merit aid programs could affect the probability of completing a bachelor's degree and alter the post–merit aid composition of college graduates. For example, if merit aid increases degree completion for marginal students and if these students are less likely to major in a STEM field, then not accounting for selection could bias the results. We might observe an erroneous decrease in the likelihood of majoring in STEM among college graduates because the merit-induced additional graduates majored in non-STEM fields. Sjoquist and Winters (forthcoming) suggest that merit programs had no significant effect on bachelor's degree completion, but the analysis in this paper uses a smaller sample (only the 2009–11 ACS), so we also examine the effects of merit programs on bachelor's degree completion using the 2009–11 ACS with the specifications in table 3. We reestimated the basic regression reported in Sjoquist and Winters (forthcoming) using the smaller sample used in the current paper; the results are reported in appendix table A2. Consistent with Sjoquist and Winters (forthcoming), the estimates are small and statistically insignificant, so selection is unlikely to be an important issue.

We next estimated column 1 of table 3 unconditional on education, that is, we redefine the sample to include all 24–39-year-olds, including those with no college degree or even no high school degree; results are in appendix table A3. The results are consistent with the results conditioned on earning a bachelor's degree or more, that is, a negative and statistically significant coefficient for the total and for males. The coefficients for the STEM variable are now smaller due to the more inclusive sample and smaller sample means, but the implied percentage decrease in STEM graduates is comparably large. For example, the coefficient and sample mean for the total population are −0.0042 and 0.0537, respectively, implying a 7.8% decrease in the number of STEM graduates uncondi-

[18] We would have preferred to use measures of student high school performance rather than inputs, but other than SAT scores, there are no measures that are consistently available by state for the time period we consider. Annual high school graduation rate estimates by state are available from Greene and Winters (2005) but only for the period 1991–2002; other than with the possible exception of New Jersey, the pattern of graduation rates is not consistent with improved education outcomes due to an educational policy.

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tional on education. We also separately estimated the baseline regression conditional on having at least a high school diploma and conditional on at least some college; the estimated decreases in the number of STEM graduates were 6.7% and 8.1%, respectively.

E. Accounting for Measurement Error in Treatment Status

Table 6 presents results that account for measurement error in treatment status. Recall that the results presented in table 3 are based on the assignment of individuals to the treatment group if they were born in a state that adopted a merit program and turned 18 after the program was implemented. However, some individuals attend high school outside their state of birth and some finish high school before or after age 18. To account for measurement error due to age when finishing high school, the first column of table 6 excludes from the sample persons who were ages 18

Table 6
Merit Program Effects on STEM Majors Accounting for Measurement Error in Treatment Status

	Excluding "Marginal" Birth Cohorts (1)	Using Probability of Living in Birth State (2)	Accounting for Both (3)
A. Total population	−.0149	−.0174	−.0187
	(.0066)**	(.0069)**	(.0084)**
	[−.0282, −.0050]	[−.0330, −.0057]	[−.0344, −.0070]
	{−7.23%}	{−8.44%}	{−9.07%}
B. Females	−.0087	−.0074	−.0106
	(.0151)	(.0119)	(.0194)
	[−.0192, .0061]	[−.0192, .0071]	[−.0231, .0038]
	{−6.44%}	{−5.47%}	{−7.84%}
C. Males	−.0234	−.0316	−.0298
	(.0109)**	(.0127)**	(.0139)**
	[−.0494, −.0073]	[−.0575, −.0154]	[−.0612, −.0098]
	{−7.85%}	{−0.60%}	{−10.00%}
State-of-birth dummies	Yes	Yes	Yes
Year-of-birth dummies	Yes	Yes	Yes
Age dummies	Yes	Yes	Yes
Sex, race/ethnicity dummies	Yes	Yes	Yes
State-of-birth trends	Yes	Yes	Yes
Strong merit states	Treatment	Treatment	Treatment
Weak merit states	Excluded	Excluded	Excluded
Non–merit states	Control	Control	Control

NOTE.—Standard errors in parentheses are clustered by state of birth. Conley-Taber 95% confidence intervals are in brackets. The numbers in braces are the coefficient divided by the corresponding pre-merit mean for the STEM dummy and represent the percentage decrease in the number of STEM graduates from the state. Column 1 excludes from the sample persons who were age 18 in the year of or year before the merit program was implemented in their birth state. Column 2 measures merit exposure by the predicted probability of going to high school in a merit state based on state of birth. See the text for further details.
** $p < .05$ based on standard errors clustered by state of birth.

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or 19 when a merit program was implemented in their birth state.[19] The earlier analysis assigns those who were 18 years old when the program was first implemented to the treatment group and those who were 19 years old to the control group. But some who were 18 when the program started could have finished high school a year earlier at age 17 and not been eligible. Similarly, some individuals who were 19 when the program started could have graduated high school at 19 and been eligible for the merit program. Column 1 of table 6 excludes these "marginal" birth cohorts from the analysis (both treatment and control groups) to reduce measurement error. The coefficient estimates are now slightly larger in magnitude (i.e., more negative) than the preferred specification in column 1 of table 3, but the difference is small.

We next account for possible measurement error in treatment assignment due to persons attending high school outside their state of birth.[20] Following Dynarski (2008), we explore measuring merit exposure based on the predicted probability of going to high school in a merit state based on state of birth. Using the sample of 15–17-year-olds in the 2000 Census and 2001–11 ACS, for each merit state we regress the probability of living in that state during high school on a complete set of state-of-birth dummies. We then use the predicted values and year of birth to compute the probability that an individual was exposed to a merit program. We then replace the merit dummy in our models with the probabilistic merit variable.[21] Results are reported in column 2 of table 6. The coefficient for the full sample increases in magnitude to −0.0174, which is somewhat larger in magnitude than that for the baseline specification in column 1 of table 3.

The third column of table 6 combines the correction procedures used for the first two columns to account for measurement error in treatment due to both year of birth and state. This increases the magnitude of the full sample coefficient to −0.0187, which suggests that strong merit aid programs decrease the number of STEM graduates by as much as 9.1%. Although the coefficient estimates accounting for measurement error are not statistically significantly different from the baseline specifications, the results suggest that the true effect of state merit aid programs may be larger than that reported in column 1 of table 3. We also reestimated the regressions reported in tables 4 and 5 correcting for measurement error in treatment status (results are not reported here); the effect on the results in tables 4 and 5 are similar to the changes observed for table 3.

[19] Results are similar excluding those who were ages 17 and 20 when the merit program in their state began.

[20] Seventy-six percent of 18-year-olds in the Census/ACS live in their birth state.

[21] The clustered standard errors in our main regressions are also robust to heteroskedasticity and therefore account for the effect of using an estimated covariate on the standard error of the estimated coefficient.

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F. Joint Probability of STEM and Living in Birth State

While the effect of merit aid on the probability of earning a STEM degree is an important policy issue, of perhaps even greater relevance to state policy makers is the effect of merit aid on the number of STEM graduates staying in their home state. Even though merit aid reduces the probability of earning a STEM degree, it may increase the probability that STEM graduates reside in the state post-college,[22] which could actually increase the number of STEM graduates in the state if the positive effect on location decisions is large enough to outweigh the negative effect on earning a STEM degree. To explore this, we estimated a regression of the joint probability that a college graduate was a STEM major and is living in their state of birth (table 7). The coefficient on merit aid is marginally statistically significant only for females, and it is negative. This implies that merit aid has a small negative effect on the likelihood that a female will both earn a degree with a STEM major and live in her state of birth. We had expected that the coefficients in table 7 would be close to zero, since the effect of merit aid on the probability of being a STEM major is negative (table 3) and Sjoquist and Winters (2014) report that merit aid has a positive effect on the probability of retention.

The statistically significant coefficient for females is unexpected. However, when we account for measurement error in treatment status, the coefficient for females decreases in magnitude to −0.0032, none of the coefficients is statistically significant, and the sign of the coefficient for the total population becomes positive (though small). Furthermore, we experimented with how the joint probability results change when excluding individual strong merit states from the analysis. For example, excluding Georgia and South Carolina reduces the joint probability coefficient for women to −0.0028, and it is not significant. This suggests that the results reported in table 7 are not stable, and thus we should not put much credence in them. We are unable to offer strong conclusions about the effect of merit aid on the joint probability of earning a STEM degree and remaining in one's state of birth.

IV. Summary and Conclusions

Since 1991 more than two dozen states have created statewide merit-based student financial aid programs. One of the main goals of these programs is usually to improve the quality of the state's workforce by increasing the number of college-educated workers in the state. However, these programs could have unintended consequences that are counter to the interests of the state. One concern in particular is that these merit aid

[22] Sjoquist and Winters (2014) find that strong merit programs increase the probability of residing in one's native state for an average college graduate, but they do not specifically examine the effects for STEM graduates.

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Table 7
Merit Effects on Joint Probability of STEM Degree and Remaining in Birth State

	Total (1)	Females (2)	Males (3)
Strong merit dummy	−.0051	−.0093	.0005
	(.0047)	(.0052)*	(.0090)
	[−.0132, .0003]	[−.0177, −.0044]	[−.0118, .0133]
Pre-merit mean in strong states	.095	.064	.135
% decrease in no. of STEM	−5.33%	−14.53%	.34%
State-of-birth dummies	Yes	Yes	Yes
Year-of-birth dummies	Yes	Yes	Yes
Age dummies	Yes	Yes	Yes
Sex, race/ethnicity dummies	Yes	Yes	Yes
State-of-birth year trends	Yes	Yes	Yes
Strong merit states	Treatment	Treatment	Treatment
Weak merit states	Excluded	Excluded	Excluded
Non–merit states	Control	Control	Control
Total no. of states	33	33	33

NOTE.—Standard errors in parentheses are clustered by state of birth. Conley-Taber 95% confidence intervals are in brackets.
* $p < .10$ based on standard errors clustered by state of birth.

programs may encourage students to alter their college curriculum. Recent years have also witnessed a growing concern in the United States that the nation's higher education system is producing an insufficient number of college graduates in the fields of science, technology, engineering, and mathematics (STEM). STEM fields are important drivers of innovation and economic growth, and US economic performance may lag behind if there are too few STEM graduates entering the workforce.

Using microdata from the American Community Survey (ACS), this paper examines whether recently adopted state merit-based financial aid programs have affected students' college major decisions. We focus on STEM fields but also investigate the effects on other majors. We also differentiate between states with strong merit programs and those with weak ones. Our results yield consistent evidence that strong state merit aid programs significantly reduce the likelihood that a young person will earn a STEM degree. This is a new and important finding. While men are more likely to major in STEM fields, our results suggest that the negative effect on STEM degrees is much larger for men than women.

Our conservative baseline estimates suggest that strong merit programs reduce the likelihood of earning a STEM degree by 1.3 percentage points, but alternative specifications suggest that the magnitude could be larger. STEM fields account for roughly 20% of all college graduates, so the baseline estimate corresponds to a 6.5% decrease in the number of STEM graduates; alternative estimates suggest the decrease could be as large as

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9.1%. Given the importance of STEM majors, this decrease is a major concern for policy makers.

Our finding that merit aid reduces the probability of earning a STEM degree could be consistent with a number of possible mechanisms discussed previously. We are unable to say much about the mechanisms driving the results, but we note the following. STEM degrees pay relatively high salaries, and the mechanisms posited by Rothstein and Rouse (2011) and Stater (2011) suggest that financial aid causes students to shift away from high-paying majors, though it is unclear how large the expected effects should be. However, business majors also have relatively high salaries, and we do not see a shift away from business.

Without knowing the mechanism through which merit aid affects the choice of major, it is hard to provide firm policy recommendations short of eliminating merit aid. If concern with obtaining the GPA necessary to maintain eligibility is the primary mechanism, there are possible changes that could be made. The GPA used to determine continuing eligibility could be based on a set of common courses that are independent of major, allowed to vary by major with a lower GPA for more challenging majors, or lowered for all majors to say a 2.0. If high school students avoid taking courses that prepare them to be STEM majors in order to become merit aid eligible, allowing merit aid eligibility to be based on the SAT or ACT score could reduce that disincentive.

Appendix

Table A1
Classification of American Community Survey Majors into Broad Categories

ACS Code	ACS Code Description	Broad Field
1100	General agriculture	Business
1101	Agriculture production and management	Business
1102	Agricultural economics	Business
1103	Animal sciences	STEM
1104	Food science	STEM
1105	Plant science and agronomy	STEM
1106	Soil science	STEM
1199	Miscellaneous agriculture	Business
1301	Environmental science	STEM
1302	Forestry	STEM
1303	Natural resources management	Business
1401	Architecture	Liberal arts
1501	Area, ethnic, and civilization studies	Liberal arts
1901	Communications	Liberal arts
1902	Journalism	Liberal arts
1903	Mass media	Liberal arts
1904	Advertising and public relations	Liberal arts
2001	Communication technologies	STEM

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Table A1 (*Continued*)

ACS Code	ACS Code Description	Broad Field
2100	Computer and information systems	STEM
2101	Computer programming and data processing	STEM
2102	Computer science	STEM
2105	Information sciences	STEM
2106	Computer information management and security	STEM
2107	Computer networking and telecommunications	STEM
2201	Cosmetology services and culinary arts	Liberal arts
2300	General education	Education
2301	Educational administration and supervision	Education
2303	School student counseling	Education
2304	Elementary education	Education
2305	Mathematics teacher education	Education
2306	Physical and health education teaching	Education
2307	Early childhood education	Education
2308	Science and computer teacher education	Education
2309	Secondary teacher education	Education
2310	Special needs education	Education
2311	Social science or history teacher education	Education
2312	Teacher education: multiple levels	Education
2313	Language and drama education	Education
2314	Art and music education	Education
2399	Miscellaneous education	Education
2400	General engineering	STEM
2401	Aerospace engineering	STEM
2402	Biological engineering	STEM
2403	Architectural engineering	STEM
2404	Biomedical engineering	STEM
2405	Chemical engineering	STEM
2406	Civil engineering	STEM
2407	Computer engineering	STEM
2408	Electrical engineering	STEM
2409	Engineering mechanics, physics, and science	STEM
2410	Environmental engineering	STEM
2411	Geological and geophysical engineering	STEM
2412	Industrial and manufacturing engineering	STEM
2413	Materials engineering and materials science	STEM
2414	Mechanical engineering	STEM
2415	Metallurgical engineering	STEM
2416	Mining and mineral engineering	STEM
2417	Naval architecture and marine engineering	STEM
2418	Nuclear engineering	STEM
2419	Petroleum engineering	STEM
2499	Miscellaneous engineering	STEM
2500	Engineering technologies	STEM
2501	Engineering and industrial management	STEM
2502	Electrical engineering technology	STEM
2503	Industrial production technologies	STEM
2504	Mechanical engineering related technologies	STEM
2599	Miscellaneous engineering technologies	STEM

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Table A1 (*Continued*)

ACS Code	ACS Code Description	Broad Field
2601	Linguistics and comparative language and literature	Liberal arts
2602	French, German, Latin, and other common foreign language studies	Liberal arts
2603	Other foreign languages	Liberal arts
2901	Family and consumer sciences	Social sciences
3201	Court reporting	Social sciences
3202	Pre-law and legal studies	Social sciences
3301	English language and literature	Liberal arts
3302	Composition and speech	Liberal arts
3401	Liberal arts	Liberal arts
3402	Humanities	Liberal arts
3501	Library science	Liberal arts
3600	Biology	STEM
3601	Biochemical sciences	STEM
3602	Botany	STEM
3603	Molecular biology	STEM
3604	Ecology	STEM
3605	Genetics	STEM
3606	Microbiology	STEM
3607	Pharmacology	STEM
3608	Physiology	STEM
3609	Zoology	STEM
3611	Neuroscience	STEM
3699	Miscellaneous biology	STEM
3700	Mathematics	STEM
3701	Applied mathematics	STEM
3702	Statistics and decision science	STEM
3801	Military technologies	STEM
4000	Interdisciplinary and multi-disciplinary studies (general)	Liberal arts
4001	Intercultural and international studies	Liberal arts
4002	Nutrition sciences	STEM
4003	Neuroscience	STEM
4005	Mathematics and computer science	STEM
4006	Cognitive science and biopsychology	STEM
4007	Interdisciplinary social sciences	Liberal arts
4008	Multi-disciplinary or general science	Liberal arts
4101	Physical fitness, parks, recreation, and leisure	Liberal arts
4801	Philosophy and religious studies	Liberal arts
4901	Theology and religious vocations	Liberal arts
5000	Physical sciences	STEM
5001	Astronomy and astrophysics	STEM
5002	Atmospheric sciences and meteorology	STEM
5003	Chemistry	STEM
5004	Geology and earth science	STEM
5005	Geosciences	STEM
5006	Oceanography	STEM
5007	Physics	STEM
5008	Materials science	STEM

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Table A1 (*Continued*)

ACS Code	ACS Code Description	Broad Field
5098	Multi-disciplinary or general science	STEM
5102	Nuclear, industrial radiology, and biological technologies	STEM
5200	Psychology	Social sciences
5201	Educational psychology	Social sciences
5202	Clinical psychology	Social sciences
5203	Counseling psychology	Social sciences
5205	Industrial and organizational psychology	Social sciences
5206	Social psychology	Social sciences
5299	Miscellaneous psychology	Social sciences
5301	Criminal justice and fire protection	Social sciences
5401	Public administration	Social sciences
5402	Public policy	Social sciences
5403	Human services and community organization	Social sciences
5404	Social work	Social sciences
5500	General social sciences	Social sciences
5501	Economics	Social sciences
5502	Anthropology and archeology	Social sciences
5503	Criminology	Social sciences
5504	Geography	Social sciences
5505	International relations	Social sciences
5506	Political science and government	Social sciences
5507	Sociology	Social sciences
5599	Miscellaneous social sciences	Social sciences
5601	Construction services	Business
5701	Electrical and mechanic repairs and technologies	Business
5801	Precision production and industrial arts	Business
5901	Transportation sciences and technologies	STEM
6000	Fine arts	Liberal arts
6001	Drama and theater arts	Liberal arts
6002	Music	Liberal arts
6003	Visual and performing arts	Liberal arts
6004	Commercial art and graphic design	Liberal Arts
6005	Film, video, and photographic arts	Liberal Arts
6006	Art history and criticism	Liberal arts
6007	Studio arts	Liberal arts
6099	Miscellaneous fine arts	Liberal arts
6100	General medical and health services	Health-related
6102	Communication disorders sciences and services	Health-related
6103	Health and medical administrative services	Health-related
6104	Medical assisting services	Health-related
6105	Medical technologies technicians	Health-related
6106	Health and medical preparatory programs	STEM
6107	Nursing	Health-related
6108	Pharmacy, pharmaceutical sciences, and administration	STEM
6109	Treatment therapy professions	Health-related
6110	Community and public health	Health-related
6199	Miscellaneous health medical professions	Health-related
6200	General business	Business

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Table A1 (*Continued*)

ACS Code	ACS Code Description	Broad Field
6201	Accounting	Business
6202	Actuarial science	STEM
6203	Business management and administration	Business
6204	Operations, logistics and e-commerce	Business
6205	Business economics	Business
6206	Marketing and marketing research	Business
6207	Finance	Business
6209	Human resources and personnel management	Business
6210	International business	Business
6211	Hospitality management	Business
6212	Management information systems and statistics	STEM
6299	Miscellaneous business and medical administration	Business
6402	History	Liberal arts
6403	United States history	Liberal arts

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Table A2
Merit Program Effects on Bachelor's Degree Completion

	(1)	(2)	(3)	(4)	(5)
A. Total population	−.0047	−.0045	−.0055	−.0059	−.0003
	(.0052)	(.0051)	(.0049)	(.0046)	(.0031)
	[−.0105, .0012]	[−.0118, .0004]	[NA]	[NA]	[−.0071, .0065]
B. Females	−.0051	−.0059	−.0069	−.0093	.0010
	(.0059)	(.0062)	(.0066)	(.0060)	(.0039)
	[−.0137, .0044]	[−.0153, .0024]	[NA]	[NA]	[−.0088, .0137]
C. Males	−.0039	−.0029	−.0038	−.0025	−.0015
	(.0071)	(.0072)	(.0066)	(.0068)	(.0046)
	[−.0141, .0052]	[−.0144, .0046]	[NA]	[NA]	[−.0113, .0089]
State-of-birth dummies	Yes	Yes	Yes	Yes	Yes
Year-of-birth dummies	Yes	Yes	Yes	Yes	Yes
Age dummies	Yes	Yes	Yes	Yes	Yes
Sex, race/ethnicity	Yes	Yes	Yes	Yes	Yes
State-of-birth × year trends	Yes	Yes	Yes	Yes	Yes
Region × year dummies	No	No	Yes	No	No
Strong merit states	Treatment	Treatment	Treatment	Treatment	Treatment
Weak merit states	Excluded	Control	Excluded	Excluded	Treatment
Non–merit states	Control	Control	Control	Excluded	Control
Total states	33	51	33	9	51

NOTE.—Standard errors in parentheses are clustered by state of birth. Conley-Taber 95% confidence intervals are in brackets; these are inestimable for cols. 3 and 4 because of how the control group is specified. NA = not applicable.

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Table A3
Merit Effects on STEM Degrees for Different Sample Restrictions by Education Level

	Education Sample Restriction			
	All (1)	High School + (2)	Some College + (3)	Bachelor's + (4)
A. Total population	−.0042	−.0040	−.0082	−.0133
	(.0018)**	(.0019)**	(.0033)**	(.0053)**
	[−.0082, −.0011]	[−.0081, −.0004]	[−.0148, −.0025]	[−.0261, −.0049]
Pre-merit mean in strong states	.054	.060	.101	.206
% decrease in no. of STEM	−7.82	−6.67	−8.12	−6.45
B. Females	−.0021	−.0022	−.0040	−.0057
	(.0031)	(.0033)	(.0050)	(.0093)
	[−.0063, .0025]	[−.0059, .0022]	[−.0098, .0024]	[−.0172, .0090]
Pre-merit mean in strong states	.039	.043	.066	.135
% decrease in no. of STEM	−5.36	−5.14	−6.02	−4.22
C. Males	−.0063	−.0059	−.0135	−.0240
	(.0026)**	(.0030)*	(.0059)**	(.0099)**
	[−.0124, −.0018]	[−.0121, −.0002]	[−.0244, −.0036]	[−.0468, −.0104]
Pre-merit mean in strong states	.069	.078	.145	.298
% decrease in no. of STEM	−9.19	−7.54	−9.30	−8.05
State-of-birth dummies	Yes	Yes	Yes	Yes
Year-of-birth dummies	Yes	Yes	Yes	Yes
Age dummies	Yes	Yes	Yes	Yes
Sex, race/ethnicity dummies	Yes	Yes	Yes	Yes
State-of-birth × year trends	Yes	Yes	Yes	Yes
Strong merit states	Treatment	Treatment	Treatment	Treatment
Weak merit states	Excluded	Excluded	Excluded	Excluded
Non–merit states	Control	Control	Control	Control

NOTE.—Standard errors in parentheses are clustered by state of birth. Conley-Taber 95% confidence intervals are in brackets.
* $p < .10$ based on standard errors clustered by state of birth.
** $p < .05$ based on standard errors clustered by state of birth.

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